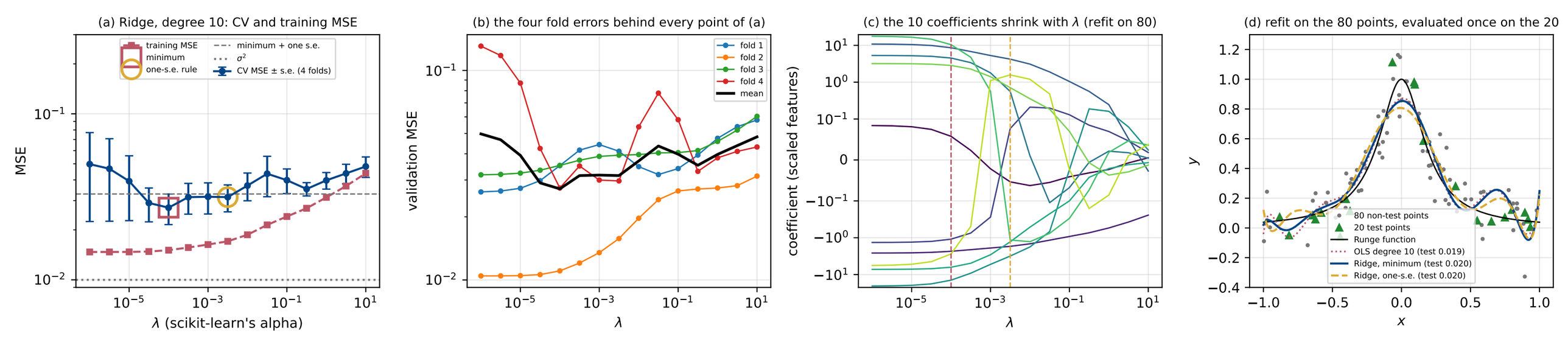
(a) Ridge, degree 10: CV and training MSE
training MSE
minimum
one-s.e. rule
minimum + one s.e.
σ^2
CV MSE ± s.e. (4 folds)
MSE
λ (scikit-learn's alpha)
(b) the four fold errors behind every point of (a)
fold 1
fold 2
fold 3
fold 4
mean
validation MSE
λ
(c) the 10 coefficients shrink with λ (refit on 80)
coefficient (scaled features)
λ
(d) refit on the 80 points, evaluated once on the 20
80 non-test points
20 test points
Runge function
OLS degree 10 (test 0.019)
Ridge, minimum (test 0.020)
Ridge, one-s.e. (test 0.020)
x
y